Single Channel Operational Amplifier

LM321

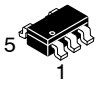
LM321 is a general purpose, single channel op amp with internal compensation and a true differential input stage. This op amp features a wide supply voltage ranging from 3 V to 32 V for single supplies and ±1.5 to ±16 V for split supplies, suiting a variety of applications. LM321 is unity gain stable even with large capacitive loads up to 1.5 nF. LM321 is available in a space-saving TSOP−5/SOT23−5 package.

Features

- Wide Supply Voltage Range: 3 V to 32 V
- Short Circuit Protected Outputs
- True Differential Input Stage
- Low Input Bias Currents
- Internally Compensated
- Single and Split Supply Operation
- Unity Gain Stable with 1.5 nF Capacitive Load
- This Device is Pb-Free, Halogen Free/BFR Free and is RoHS Compliant

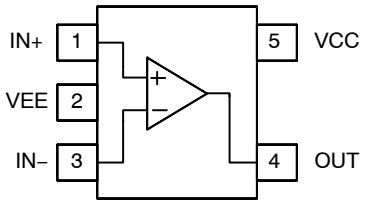
Typical Applications

- Gain Stage
- Active Filter
- Signal Processing

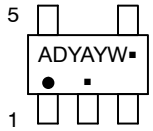


TSOP−5
CASE 483

PIN CONNECTION



MARKING DIAGRAM



ADY = Specific Device Code
A = Assembly Location
Y = Year
W = Work Week
▪ = Pb-Free Package
(Note: Microdot may be in either location)

ORDERING INFORMATION

Device	Package	Shipping†
LM321SN3T1G	TSOP−5 (Pb−Free)	3000 / Tape & Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

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Table 1. ABSOLUTE MAXIMUM RATINGS (Over operating free-air temperature, unless otherwise stated)

Parameter	Rating	Unit
Supply Voltage	36	V
INPUT AND OUTPUT PINS		
Input Voltage	V_{EE} – 0.3 to 32	V
Input Current	±10	mA
Output Short Circuit Duration (Note 1)	Continuous	
TEMPERATURE		
Operating Temperature	–40 to +125	°C
Storage Temperature	–65 to +150	°C
Junction Temperature	–65 to +150	°C
ESD RATINGS (Note 2)		
Human Body Model (HBM)	200	V
Charged Device Model (CDM)	800	V
Machine Model (MM)	100	V
OTHER RATINGS		
Latch-Up Current (Note 3)	100	mA
MSL	Level 1	

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. Short circuits can cause excessive heating and eventual destruction.
2. This device series incorporates ESD protection and is tested by the following methods:
   ESD Human Body Model tested per JEDEC standard: JESD22–A114
   ESD Machine Model tested per JEDEC standard: JESD22–A115
3. Latch-up Current tested per JEDEC standard: JESD78

Table 2. THERMAL INFORMATION (Note 4)

Parameter	Symbol	Package	Value	Unit
Junction to Ambient	θ_{JA}	TSOP–5/SOT23–5	235	°C/W

4. As mounted on an $80 \times 80 \times 1.5$ mm FR4 PCB with 650 mm^2 and 2 oz (0.034 mm) thick copper heat spreader. Following JEDEC JESD/EIA 51.1, 51.2, 51.3 test guidelines.

Table 3. RECOMMENDED OPERATING CONDITIONS

Parameter	Symbol	Range	Unit
Supply Voltage (V_{CC} – V_{EE})	V_S	3 to 32	V
Specified Operating Range	T_A	–40 to 85	°C
Common Mode Input Voltage Range	V_{CM}	V_{EE} to V_{CC}–1.7	V

Functional operation above the stresses listed in the Recommended Operating Ranges is not implied. Extended exposure to stresses beyond the Recommended Operating Ranges limits may affect device reliability.

Table 4. ELECTRICAL CHARACTERISTICS – V_S = 5 V

(At T_A = +25°C, R_L = 10 kΩ connected to mid-supply, V_{CM} = V_{OUT} = mid-supply, unless otherwise noted.
Boldface limits apply over the specified temperature range, T_A = –40°C to 85°C, guaranteed by characterization and/or design.)

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
INPUT CHARACTERISTICS						
Offset Voltage	V_{OS}	V_S = 5 V, V_{CM} = V_{EE} to V_{CC} – 1.7 V T_A = 25°C T_A = –40°C to 85°C	 – –	 0.3 –	 7 **9**	mV
Offset Voltage Drift vs Temp	$\Delta V_{OS}/\Delta T$	T_A = –40°C to 85°C	–	**7**	–	μV/°C
Input Bias Current	I_{IB}	T_A = 25°C T_A = –40°C to 85°C	– –	–10 –	– **–500**	nA
Input Offset Current	I_{OS}	T_A = 25°C T_A = –40°C to 85°C	– –	1 –	– **150**	nA
Common Mode Rejection Ratio	CMRR	V_{CM} = V_{EE} to V_{CC} – 1.7 V	65	85	–	dB
Input Resistance	R_{IN}	Differential Common Mode	– –	85 300	– –	GΩ
Input Capacitance	C_{IN}	Differential Common Mode	– –	0.6 1.6	– –	pF
OUTPUT CHARACTERISTICS						
Open Loop Voltage Gain	A_{VOL}		–	100	–	dB
Open Loop Output Impedance	Z_{OUT_OL}	f = UGBW, I_O = 0 mA	–	1,200	–	Ω
Output Voltage High	V_{OH}	R_L = 2 kΩ to V_{EE} R_L = 10 kΩ to V_{EE}	**V_{CC}–1.8** **V_{CC}–1.8**	V_{CC}–1.4 V_{CC}–1.4	– –	V
Output Voltage Low	V_{OL}	R_L = 10 kΩ to V_{CC}	–	V_{EE}+0.8	**V_{EE}+1.0**	V
Output Current Capability	I_O	Sinking Current V_S = 5 V V_S = 15 V	 **10** **10**	 20 20	 – –	mA
Output Current Capability	I_O	Sourcing Current V_S = 5 V V_S = 15 V	 **20** **20**	 40 40	 – –	mA
Capacitive Load Drive	C_L	Phase Margin = 15°	–	1,500	–	pF
NOISE PERFORMANCE						
Voltage Noise Density	e_N	f_{IN} = 1 kHz	–	40	–	nV/√Hz
DYNAMIC PERFORMANCE						
Gain Bandwidth Product	GBWP	C_L = 25 pF, R_L to V_{CC}	–	750	–	kHz
Gain Margin	A_M	C_L = 25 pF, R_L to V_{CC}	–	14	–	dB
Phase Margin	α_M	C_L = 25 pF, R_L to V_{CC}	–	60	–	°
Slew Rate	SR	C_L = 25 pF, R_L = ∞	–	0.3	–	V/μs
POWER SUPPLY						
Power Supply Rejection Ratio	PSRR	V_S = 5 V to 32 V	**62**	100	–	dB
Quiescent Current	I_Q	No Load	–	0.25	**0.5**	mA

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

Table 5. ELECTRICAL CHARACTERISTICS – V_S = 32 V

(At T_A = +25°C, R_L = 10 kΩ connected to mid-supply, V_{CM} = V_{OUT} = mid-supply, unless otherwise noted.
Boldface limits apply over the specified temperature range, T_A = –40°C to 85°C, guaranteed by characterization and/or design.)

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
INPUT CHARACTERISTICS						
Offset Voltage	V_{OS}	V_S = 32 V, V_{CM} = V_{EE} to V_{CC} – 1.7 V T_A = 25°C T_A = –40°C to 85°C	 – –	 0.3 –	 7 **9**	mV
Offset Voltage Drift vs Temp	$\Delta V_{OS}/\Delta T$	T_A = –40°C to 85°C	–	**7**	–	μV/°C
Common Mode Rejection Ratio	CMRR	V_{CM} = V_{EE} to V_{CC} – 1.7 V	–	100	–	dB
OUTPUT CHARACTERISTICS						
Open Loop Voltage Gain	A_{VOL}	T_A = 25°C T_A = –40°C to 85°C	– **84**	100 –	– –	dB
Open Loop Output Impedance	Z_{OUT_OL}	f = UGBW, I_O = 0 mA	–	2,000	–	Ω
Output Voltage High	V_{OH}	R_L = 2 kΩ to V_{EE} R_L = 10 kΩ to V_{EE}	**V_{CC}–2.5** **V_{CC}–2.5**	V_{CC}–2.0 V_{CC}–1.5	– –	V
Output Voltage Low	V_{OL}	R_L = 10 kΩ to V_{CC}	–	V_{EE}+1.0	**V_{EE}+1.5**	V
Capacitive Load Drive	C_L	Phase Margin = 15°	–	1,500	–	pF
NOISE PERFORMANCE						
Voltage Noise Density	e_N	f_{IN} = 1 kHz	–	40	–	nV/√Hz
Total Harmonic Distortion + Noise	THD+N	V_S = 30 V, f_{IN} = 1 kHz, R_L to V_{CC}	–	0.02	–	%
DYNAMIC PERFORMANCE						
Gain Bandwidth Product	GBWP	C_L = 25 pF, R_L to V_{CC}	–	900	–	kHz
Gain Margin	A_M	C_L = 25 pF, R_L to V_{CC}	–	18	–	dB
Phase Margin	α_M	C_L = 25 pF, R_L to V_{CC}	–	66	–	°
Slew Rate	SR	C_L = 25 pF, R_L = ∞	–	0.4	–	V/μs
POWER SUPPLY						
Power Supply Rejection Ratio	PSRR	V_S = 5 V to 32 V	**62**	100	–	dB
Quiescent Current	I_Q	No Load, V_S = 32 V	–	0.3	**1.2**	mA

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

TYPICAL CHARACTERISTICS

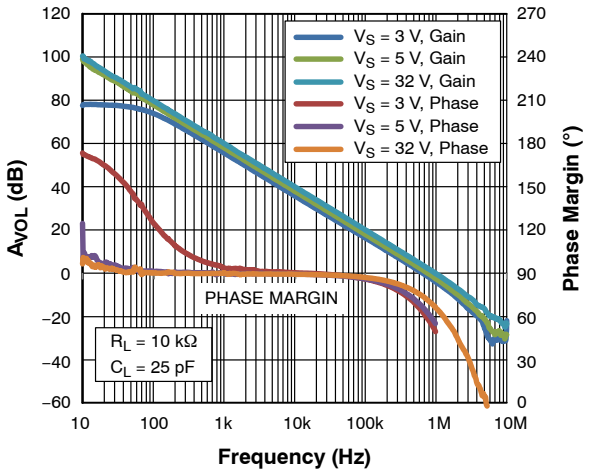


Figure 1. Open Loop Gain and Phase Margin vs. Frequency

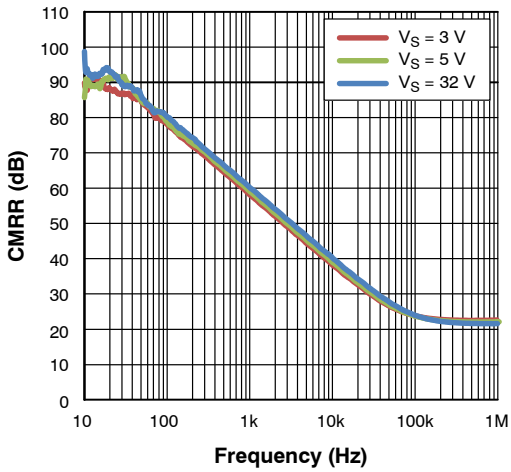


Figure 2. CMRR vs. Frequency

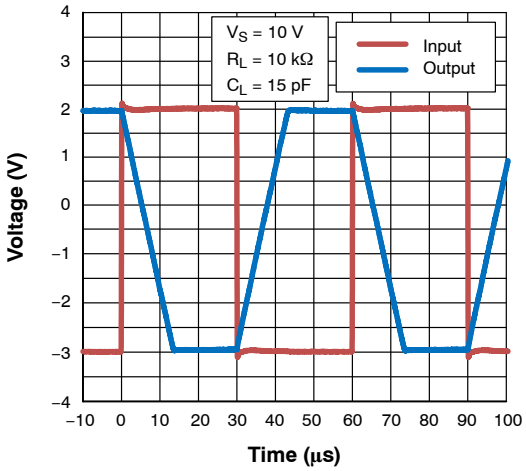


Figure 3. Inverting Large Signal Step Response

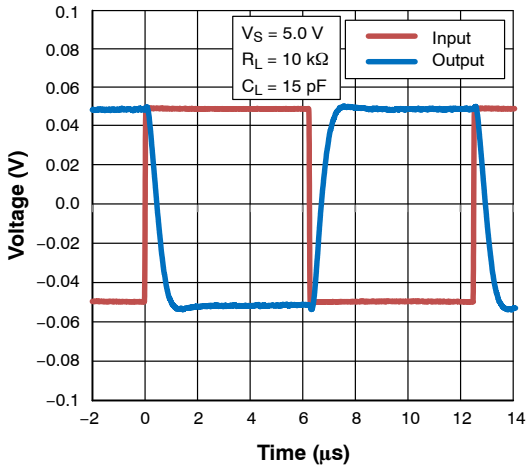


Figure 4. Inverting Small Signal Step Response

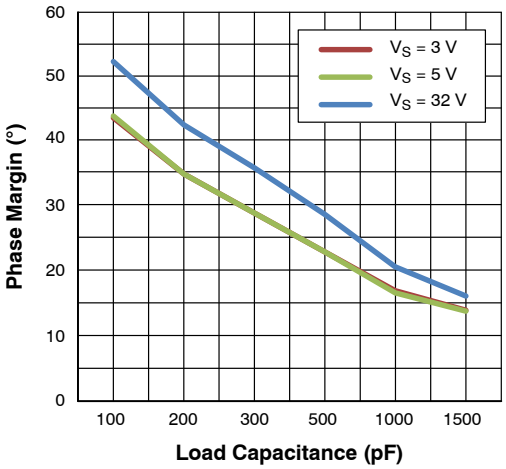


Figure 5. Phase Margin vs. Load Capacitance

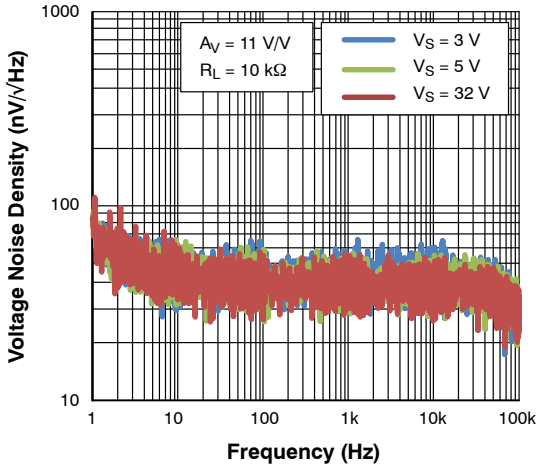


Figure 6. Voltage Noise Density vs. Frequency

TYPICAL CHARACTERISTICS

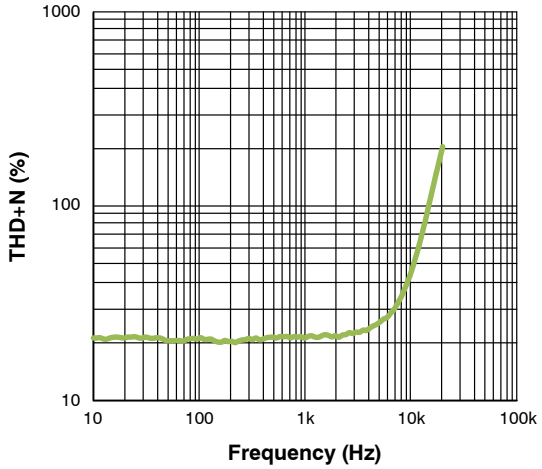


Figure 7. THD+N vs. Frequency

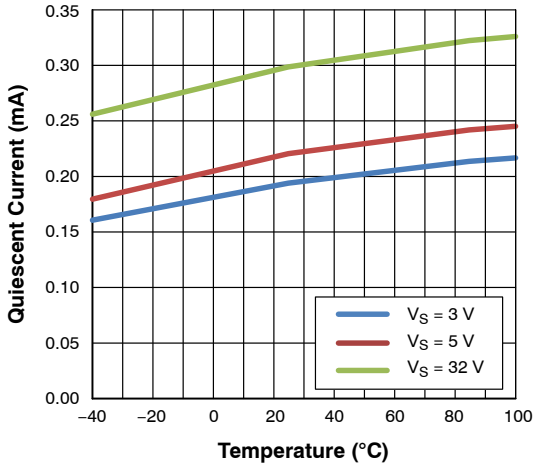


Figure 8. Quiescent Current vs. Temperature

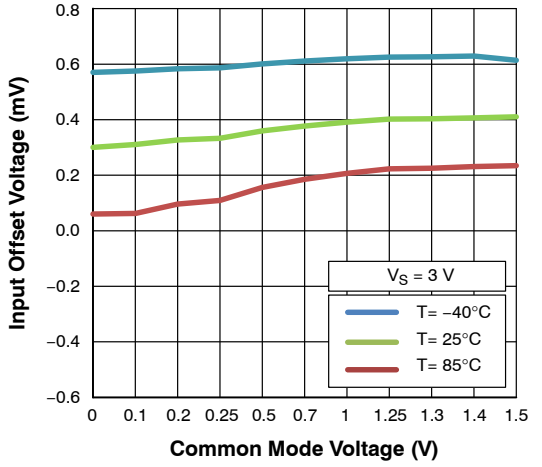


Figure 9. Input Offset Voltage vs. Common Mode Voltage at 3 V Supply

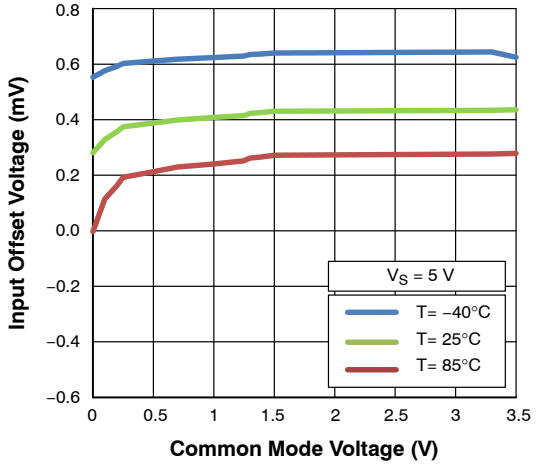


Figure 10. Input Offset Voltage vs. Common Mode Voltage at 5 V Supply

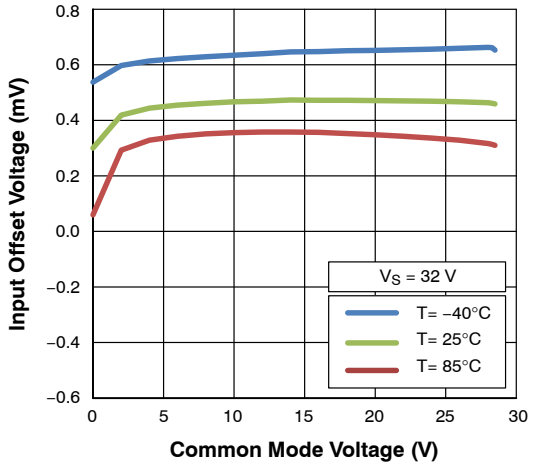


Figure 11. Input Offset Voltage vs. Common Mode Voltage at 32 V Supply

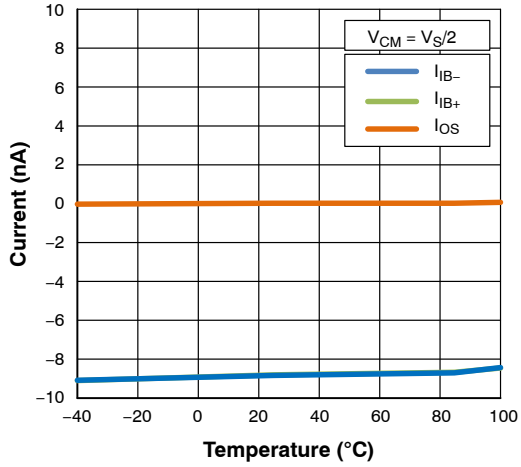


Figure 12. Input Bias and Offset Current vs. Temperature

TYPICAL CHARACTERISTICS

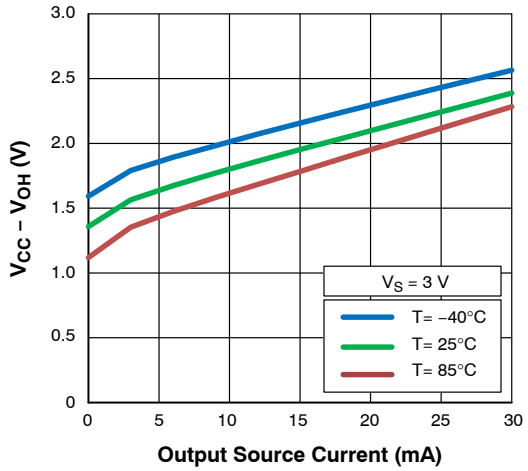


Figure 13. High Level Output Voltage Swing vs. Output Current at 3 V Supply

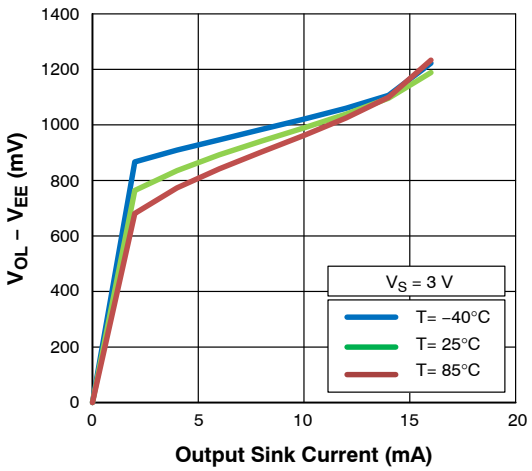


Figure 14. Low Level Output Voltage Swing vs. Output Current at 3 V Supply

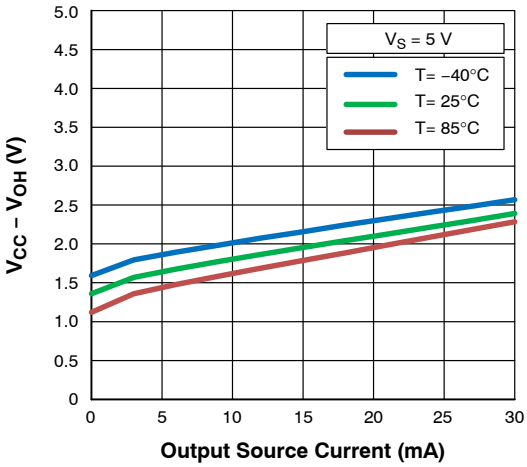


Figure 15. High Level Output Voltage Swing vs. Output Current at 5 V Supply

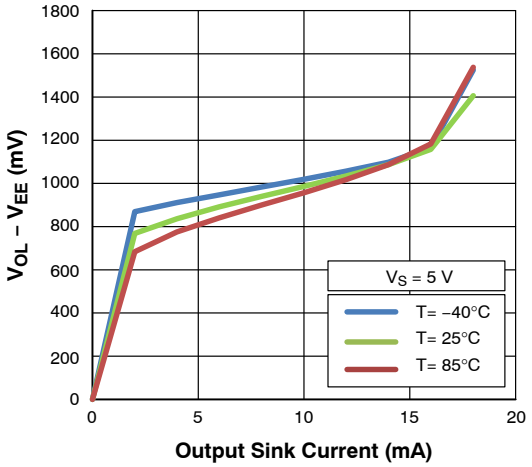


Figure 16. Low Level Output Voltage Swing vs. Output Current at 5 V Supply

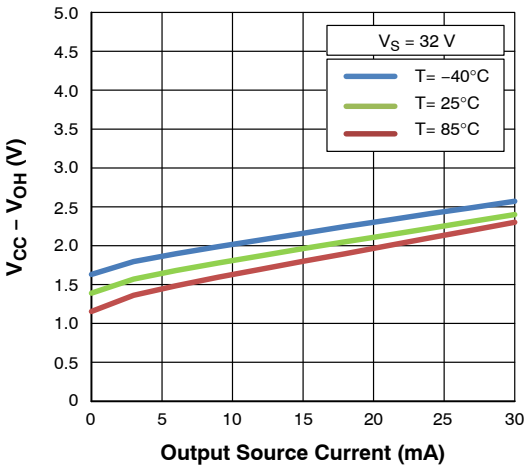


Figure 17. High Level Output Voltage Swing vs. Output Current at 32 V Supply

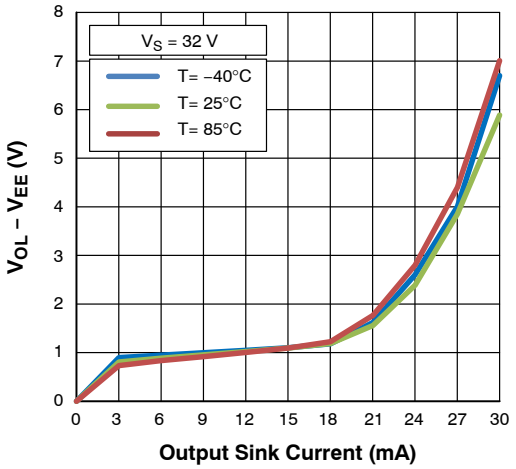


Figure 18. Low Level Output Voltage Swing vs. Output Current at 32 V Supply

APPLICATION INFORMATION

CIRCUIT DESCRIPTION

The LM321 is made using two internally compensated, two−stage operational amplifiers. The first stage of each consists of differential input devices Q20 and Q18 with input buffer transistors Q21 and Q17 and the differential to single ended converter Q3 and Q4. The first stage performs not only the first stage gain function but also performs the level shifting and transconductance reduction functions. By reducing the transconductance, a smaller compensation capacitor (only 5.0 pF) can be employed, thus saving chip area. The transconductance reduction is accomplished by splitting the collectors of Q20 and Q18. Another feature of this input stage is that the input common mode range can include the negative supply or ground, in single supply operation, without saturating either the input devices or the differential to single−ended converter. The second stage consists of a standard current source load amplifier stage.

Each amplifier is biased from an internal−voltage regulator which has a low temperature coefficient thus giving each amplifier good temperature characteristics as well as excellent power supply rejection.

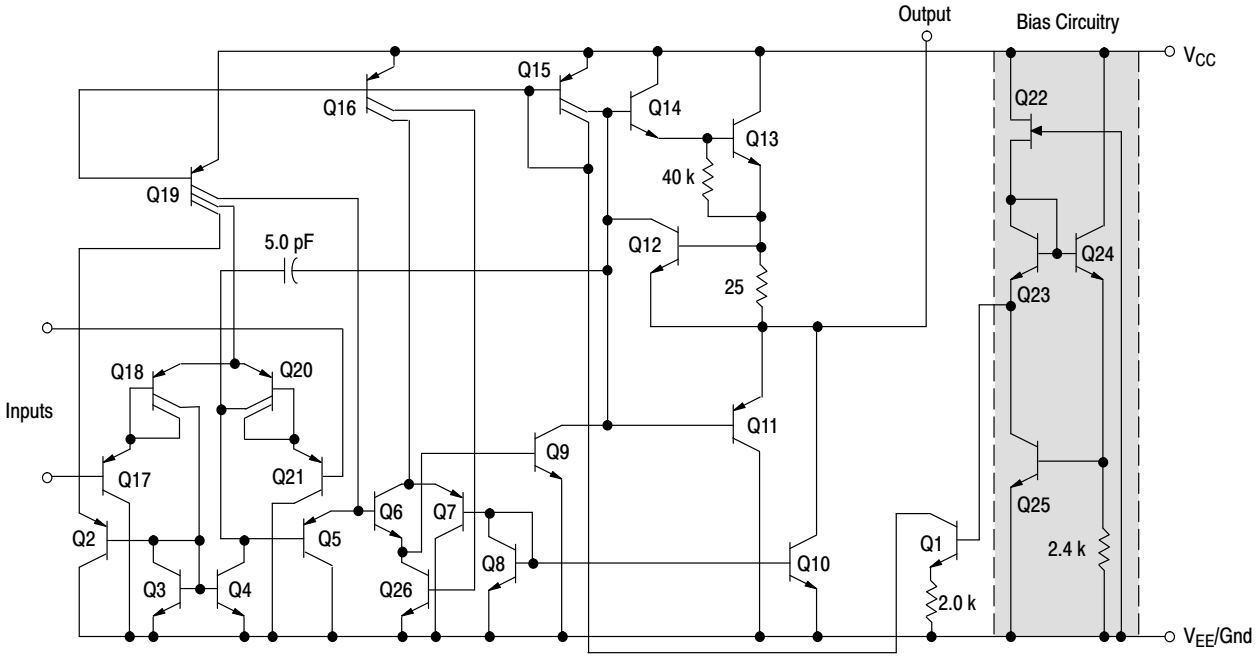


Figure 19. LM321 Representative Schematic Diagram

LM321 has a class B output stage, which is comprised of push−pull transistors. This type of output is inherently subject to crossover distortion near mid−rail where neither push or pull transistors are conducting. Several techniques can be used to minimize crossover distortion. Connecting the output load to either the positive or negative supply rail instead of mid−rail can reduce the crossover distortion. Additionally, increasing the load resistance relatively decreases the amount of crossover distortion.

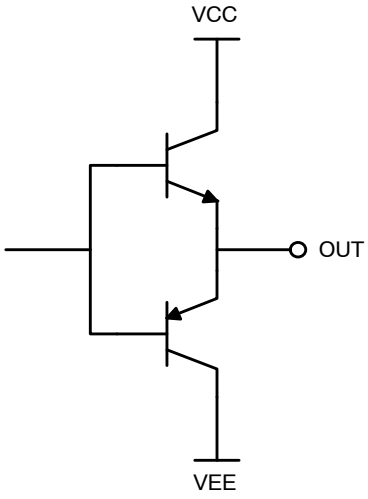


Figure 20. Simplified Class B Output

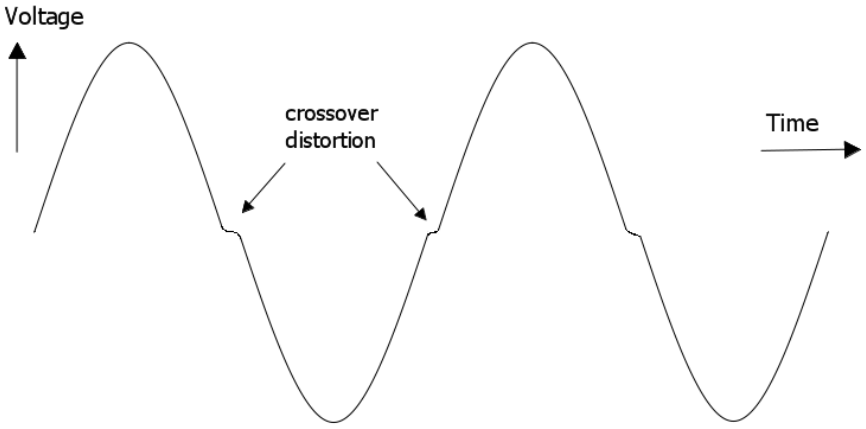
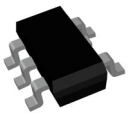
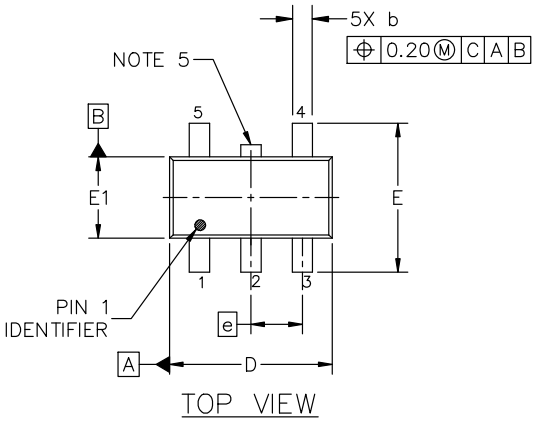


Figure 21. Sine wave with crossover distortion

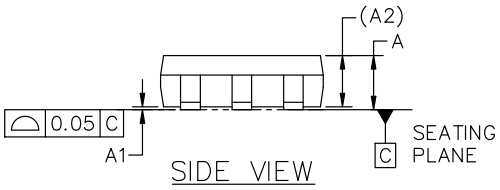


TSOP−5 3.00x1.50x0.95, 0.95P
CASE 483
ISSUE P

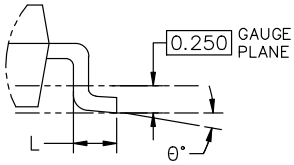
DATE 01 APR 2024



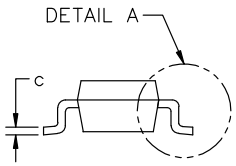
TOP VIEW



SIDE VIEW



DETAIL "A"
SCALE 2:1



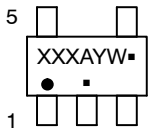
END VIEW

NOTES:

1. DIMENSIONING AND TOLERANCING CONFORM TO ASME Y14.5−2018.
2. ALL DIMENSION ARE IN MILLIMETERS (ANGLES IN DEGREES).
3. MAXIMUM LEAD THICKNESS INCLUDES LEAD FINISH THICKNESS. MINIMUM LEAD THICKNESS IS THE MINIMUM THICKNESS OF BASE MATERIAL.
4. DIMENSIONS D AND E1 DO NOT INCLUDE MOLD FLASH, PROTRUSIONS OF GATE BURRS. MOLD FLASH, PROTRUSIONS, OR GATE BURRS SHALL NOT EXCEED 0.15 PER SIDE. DIMENSION D.
5. OPTIONAL CONSTRUCTION: AN ADDITIONAL TRIMMED LEAD IS ALLOWED IN THIS LOCATION. TRIMMED LEAD NOT TO EXTEND MORE THAN 0.2 FROM BODY.

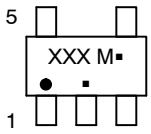
DIM	MILLIMETERS		
	MIN.	NOM.	MAX.
A	0.900	1.000	1.100
A1	0.010	0.055	0.100
A2		0.950 REF.	
b	0.250	0.375	0.500
c	0.100	0.180	0.260
D	2.850	3.000	3.150
E	2.500	2.750	3.000
E1	1.350	1.500	1.650
e		0.950 BSC	
L	0.200	0.400	0.600
θ	0°	5°	10°

GENERIC
MARKING DIAGRAM*



Analog

XXX = Specific Device Code
A = Assembly Location
Y = Year
W = Work Week
▪ = Pb−Free Package

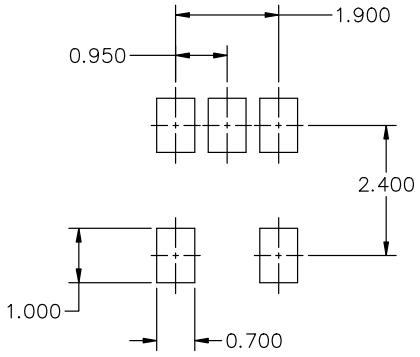


Discrete/Logic

XXX = Specific Device Code
M = Date Code
▪ = Pb−Free Package

(Note: Microdot may be in either location)

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.



RECOMMENDED MOUNTING FOOTPRINT*

* FOR ADDITIONAL INFORMATION ON OUR Pb−FREE STRATEGY AND SOLDERING DETAILS, PLEASE DOWNLOAD THE ON SEMICONDUCTOR SOLDERING AND MOUNTING TECHNIQUES REFERENCE MANUAL, SOLDERRM/D.

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